

Editor-in-Chief

Suranaree Journal of Science and Technology

Dear Editor,

With great pleasure, we submit our original work, presented at ***“ICAPMA-ICREM 2025, Pattaya, Thailand,”*** for your review and publication in SUJST, titled **“OPTIMIZATION OF DIGITAL HOLOGRAPHY WITH DEEP LEARNING FOR NON-DESTRUCTIVE DEFECT DETECTION IN PHOTOVOLTAIC CELLS.”** At the Extreme Conditions *AI Photonics, Advanced Research Laboratory, Departments of Physics, School of Science, King Mongkut's Institute of Technology Ladkrabang, Bangkok 10520, Thailand.* **Abdullahi Bako, Suebtarkul Suchat, and Prathan Buranasiri** carried out the study.

This study presents a deep learning aligned with a digital holography framework for non-contact surface defect detection in photovoltaic (PV) cells. An automated workflow was employed to train the YOLOv8 model via the Roboflow platform. The YOLOv8 object detection model was developed to analyze 800 annotated holographic images of PV cells acquired in an indoor setup. The model effectively identifies subtle defects. A roughness defect was identified in the analysis, with a clear distinction between affected and unaffected areas, compared to other traditional inspection/testing techniques like EL, PL, AFM, profilometry, and infrared thermography.

The manuscript is original, has not been published elsewhere, and is not under consideration by another journal. We respectfully request your consideration for peer review.

Please find the manuscript attached. We appreciate your consideration of our manuscript for publication and look forward to the possibility of contributing to the SUJST Publishing Group.

Thank you for your time and consideration. We look forward to your response.

Sincerely,

Abdullahi Bako

On behalf of all co-authors

King Mongkut's Institute of Technology Ladkrabang, Bangkok 10520, Thailand.